

# LINFO1115 Exam August 2022

TOTAL POINTS

**28 / 30**

## QUESTION 1

### Structural balance 5 pts

#### 1.1 Definition of structural balance 2 / 2

- ✓ - **0 pts** Correct
- **0.5 pts** Missing explanation that graph is complete.
- **1 pts** Wrong triangle

#### 1.2 Structural balance theorem 2 / 3

- **0 pts** Correct
- **2 pts** Proof incorrect
- ✓ - **1 pts** Incorrect or missing reasoning w.r.t. X & Y
- **0.25 pts** Missing case when only one camp
- **1 pts** Statement of theorem is missing or incorrect
- **0.5 pts** Statement of theorem is partially correct
- **3 pts** Incorrect

## QUESTION 2

### Second-price auction theorem 5 pts

#### 2.1 Second-price auction in the language of game theory 2 / 2

- ✓ - **0 pts** Correct
- **1 pts** Incorrect or missing payoff when bidder i wins
- **1 pts** Incorrect or missing payoff when bidder i loses
- **1 pts** Explanation has errors
- **2 pts** Incorrect

#### 2.2 Second-price auction theorem 3 / 3

- ✓ - **0 pts** Correct
- **1 pts** Some errors or missing steps in explanation but mainly right
- **3 pts** Incorrect

- **2 pts** Missing explanation when bidder i's win status flips (from win to lose, or from lose to win)

## QUESTION 3

### 3 Trading networks with intermediaries 4 / 4

- ✓ - **0 pts** Correct
- **1 pts** Missing equilibrium for monopoly (trader wins all, buyer and seller have zero)
- **1 pts** Missing or wrong equilibrium for competition (buyer and seller split the gains, trader has zero)
- **2 pts** Incorrect for perfect competition

## QUESTION 4

### 4 Bow-tie structure of the Web 3 / 3

- ✓ - **0 pts** Correct
- **0.5 pts** Missing tubes
- **3 pts** Incorrect
- **1 pts** Missing relative sizes

## QUESTION 5

### 5 Bayes' Rule 2 / 3

- **0 pts** Correct
- **3 pts** Incorrect
- **2 pts** Incorrect Bayes' Rule for computation
- ✓ - **1 pts** Correct Bayes' Rule but error in computation

## QUESTION 6

### 6 The Long Tail 2 / 2

- ✓ - **0 pts** Correct
- **2 pts** Incorrect
- **1.5 pts** Wrong explanation with a kernel of truth
- **1 pts** Partial explanation

## QUESTION 7

### Cascading in a network 8 pts

## 7.1 Cascade diffusion model 4 / 4

✓ - 0 pts Correct

- 1 pts Missing explanation for edge game
- 1 pts Missing explanation for neighbor threshold
- 4 pts Incorrect
- 2 pts Incorrect edge game
- 2 pts Incorrect neighbor threshold

## 7.2 The cluster-cascade theorem 4 / 4

✓ - 0 pts Correct

- 4 pts Incorrect
- 2 pts One direction missing
- 1 pts One direction only has example

# LINFO1115 Final Exam

August 17, 2022

(Note: Q1 and Q2 taken together correspond to the midterm.)

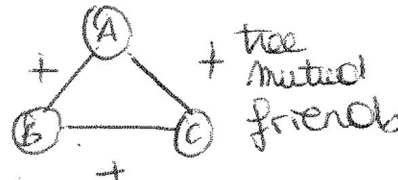
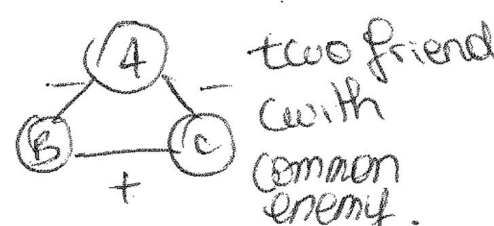
## Q1 Structural balance.

### Q1.1 Definition of structural balance. [2pts]

Give a precise definition when a graph is balanced. Don't forget to say for what graphs balance is defined.

a labeled complete graph is balanced if for each triangle in the graph, the links are  $(+++)$  or  $(+--)$  (triple of 3 nodes)

Balanced

### Q1.2 Structural balance theorem. [3pts]

The balance theorem makes a connection between a local property and a global property for a graph. State the balance theorem and give a proof.

if a labeled complete graph is balanced, it means that all pairs of nodes are friends, or nodes can be divided into two groups such that nodes between a group are friends and nodes between groups are enemies.

~~we suppose~~ if the graph has no negative links the proof is done

on definit  $X$  et  $Y$ , each sets consists of mutual friend, and the sets are enemies.

define set  $X$  to be  $A$  plus all of its friends

define set  $Y$  to be all the members of  $A$

we need to prove that

Every two nodes in  $X$  are friends, Every two nodes in  $Y$  are friends, Every node in  $X$  is enemy of every node  $Y$

First and last name [REDACTED]

NOMA [REDACTED]

## Q2 Second-price auction theorem.

### Q2.1 Second-price auction in the language of game theory. [2pts]

Assume each bidder  $i$  is a player and bidder  $i$ 's strategy is the bid amount  $b_i$  as function of  $i$ 's true value  $v_i$ . A game is played when each bidder bids once; one bidder wins and the others lose. In your answer, define the payoff to each bidder  $i$ .

define as follow:

if  $b_i$  is not the winning bid, then the payoff to  $i$  is 0. if  $b_i$  is the winning bid, and some other  $b_j$  is the second-place bid - then the payoff would be for  $i$  ( $v_i - b_j$ )

### Q2.2 Second-price auction theorem. [3pts]

We proved the theorem that in a sealed second-price auction, it is a dominant strategy for each bidder to choose a bid  $b_i$  equal to the bidder's true value  $v_i$ . To prove the theorem, we assume that each bidder bids  $b_i = v_i$  and we show that no deviation from this will improve the payoff. The proof considers two cases: (1)  $i$  raises bid, and (2)  $i$  lowers bid. For this question, choose one of these two cases and explain precisely why  $i$ 's payoff will not improve in that case. You can make a diagram, as done in the course.

par les deux cas, la somme qui devra être payé dépendant de l'autre bids, le plus large

on suppose bidder  $i$  choisit  $b_i' > v_i$

- ça affecte le payoff si il perd avec  $v_i$  et gagne avec  $b_i$

- second-place bid  $b_i$  doit être entre  $b_i$  et  $b_i'$

payoff to  $i$  would be at most  $v_i - b_i \leq 0$

↑ higher bid

Alternate bid  $b_i'$  not improved.

↑ truthful bid  $b_i = v_i$

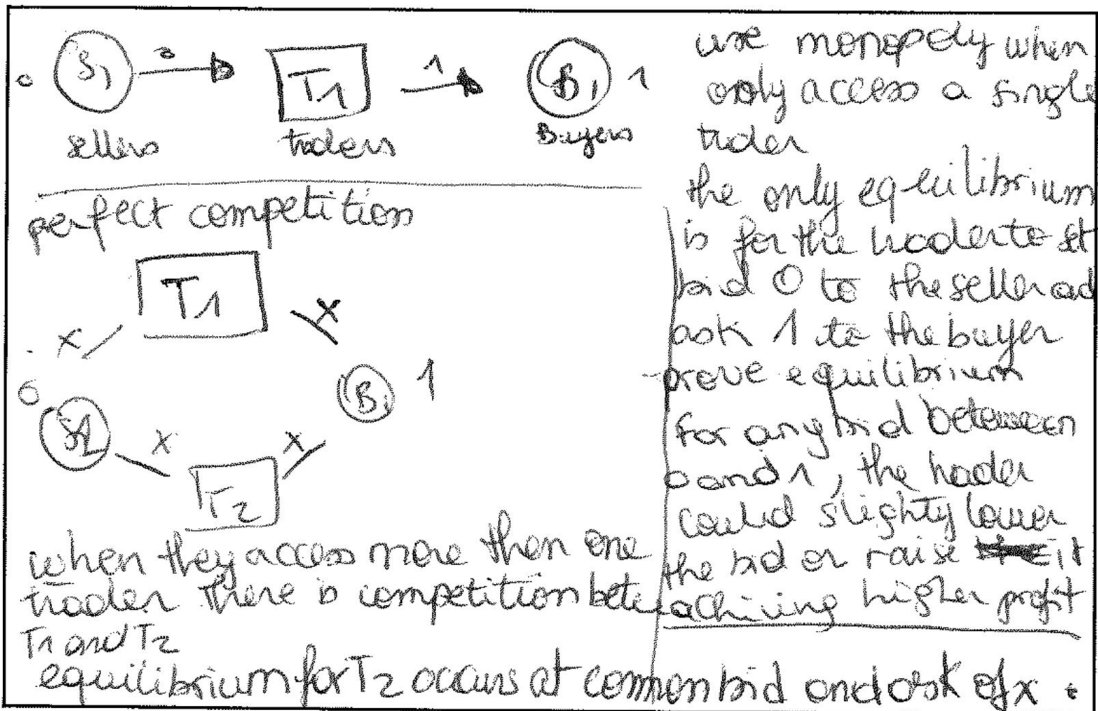
↑ Alternative bid  $b_i''$

↓ lower bid.

|                     |            |
|---------------------|------------|
| First and last name | [REDACTED] |
| NOMA                | [REDACTED] |

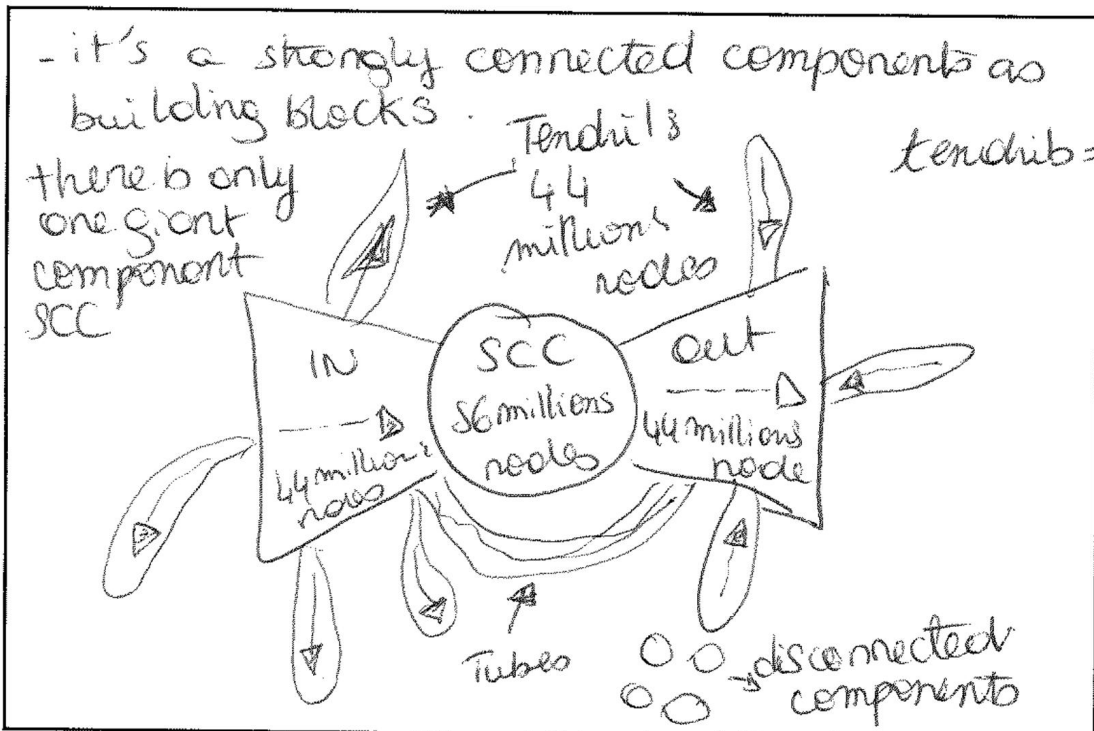
### Q3 Trading networks with intermediaries. [4pts]

We studied the strategies of nodes on a trading network, which depend on the topology of the network. We saw two building blocks, namely monopoly and perfect competition. Define monopoly, give an example of a trading network with a monopoly, and give the equilibrium bids for your network. Define perfect competition, give an example of a trading network with perfect competition, and give the equilibrium bids for your network.



### Q4 Bow-tie structure of the Web. [3pts]

Explain in detail the bow-tie structure of the Web, including the relative sizes of the different parts.



First and last name: [redacted]

NOMA [redacted]

**Q5 Bayes' Rule. [3pts]**

In a particular city, 90% of taxi cabs are black and 10% are yellow. A witness to an accident says the taxi was yellow. Assume witnesses sometimes misidentify colors of cabs: the color they say is only correct 75% of the time. Use Bayes' Rule to compute the probability that the cab is yellow, if the witness says it is yellow. Show the computation.

$$Pr[true = Y | report = Y] = \frac{Pr[true = Y] \times Pr[report = Y | true = Y]}{Pr[report = Y]}$$

$$Pr[report = Y | true = Y] = 0.75, Pr[true = Y] = 0.1$$

$$Pr[report = Y] = 0.1 \cdot 0.75 + 0.9 \cdot 0.25 = 0.325$$

$$Pr[true = Y | report = Y] = 0.1 \cdot (0.75 / 0.325) = 0.23$$

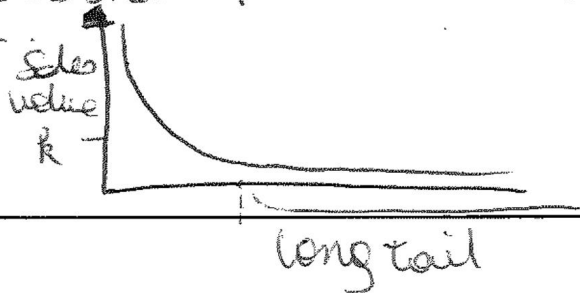
the cab is yellow with 23% probability

**Q6 The Long Tail. [2pts]**

Define the Long Tail and show a graph. Explain how the Long Tail has helped Amazon become a Web commerce giant.

for Amazon the ability to carry huge inventories is a big reason for their success, this much easier to implement on a web site than in a brick-and-mortar store.

long tail  $\Rightarrow$  Power laws mean that the popularity decreases very slowly  
 - a huge number of unpopular pages can have big aggregate popularity  
 - success of many web sites depends on having a huge inventory



First and last name

NOMA

## Q7 Cascading in a network.

### Q7.1 Cascade diffusion model. [4pts]

Node decisions propagate through networks by following the network structure. Explain how this propagation works in two steps. First, define each edge as a coordination game. Second, given a node with a set of neighbors, define the condition under which the node will adopt a new choice.

cascading needed = All not use these behavior  
small set use A, the remaining nodes evaluate the  
payoff according to the coordination game. some  
neighbors might decide to switch to A ~~and~~  
~~so further~~

each edge  $(v, w)$   
the edge gives them an  
incentive for their matching  
behavior

|   | A      | B      |
|---|--------|--------|
| A | $a, a$ | $a, 0$ |
| B | $0, 0$ | $b, b$ |

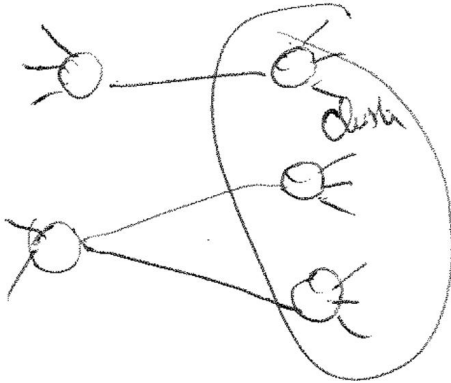
$P \geq \frac{b}{a+b}$

### Q7.2 The cluster-cascade theorem. [4pts]

We saw that diffusion can stop when it encounters a tightly knit community, called a **cluster**. A cluster of density  $p$  is a set of nodes such that each node has at least a fraction  $p$  of its neighbors in the set. We saw the cluster-cascade theorem: given a set of initial adopters with threshold  $q$ , then a complete cascade is blocked if and only if the remaining network has a cluster of density greater than  $(1-q)$ . Give a proof of this theorem. Hint: prove each direction separately.

Consider an arbitrary network, Suppose that network  
has a cluster of density  $> (1-q)$

- Assume the opposite  $v$  has adopted A
- Only neighbors of  $v$  using A are outside  
but since the cluster density  $> (1-q)$  therefore  
less than fraction  $q$  of  $v$ 's neighbors  
are outside the cluster



consider any node  $w$  in  $S$   
since  $w$  does not switch  
using A is less than  $q$ .

therefore the fraction  $b/a > (1-q)$   
therefore  $S$  is a cluster of density  
 $(1-q)$

nodes that  
initially  
adopters

nodes that don't  
eventually switch